Challenge 2: Excluded Minors of $\mathrm{GF}(p^r)$-representable matroids

Definition 1 (Matroid)**.** A *matroid* is a pair $M = (E, \mathcal{I})$ consisting of a finite set E and a collection $\mathcal{I}$ of subsets of E, called the *independent sets*, such that:

1. $\emptyset \in \mathcal{I}$;
2. if $I \in \mathcal{I}$ and $J \subseteq I$, then $J \in \mathcal{I}$;
3. if $I, J \in \mathcal{I}$ and $|I| < |J|$, then there is some $x \in J \setminus I$ such that $I \cup \{x\} \in \mathcal{I}$.

The maximal independent sets of M are called the *bases* of M.

Definition 2 ($\mathrm{GF}(p^r)$-representable matroid)**.** Let p be a prime number. Let r be a positive integer. Let $\mathrm{GF}(p^r)$ denote the finite field with p^r elements. A matroid M is $\mathrm{GF}(p^r)$-*representable* if there is a matrix A over $\mathrm{GF}(p^r)$ such that the independent sets of M are exactly the sets of columns of A which are linearly independent over $\mathrm{GF}(p^r)$.

Definition 3 (Matroid minor)**.** Let M be a matroid with ground set E. The *dual* of M, denoted M^*, is the matroid with the same ground set, where the bases are the complements of the bases of M. Given a subset $D \subseteq E$, the *deletion* of D from M, denoted $M \setminus D$, is the restriction of M to $E \setminus D$. Given a subset $C \subseteq E$, the *contraction* of M onto $E - C$ is the matroid $M/C := (M^* \setminus C)^*$, with ground set $E \setminus C$.

Given matroids M, N, then N is a *minor* of M if there exist sets C and D such that $N = M/C \setminus D$.

Definition 4 (Excluded minor)**.** Let $\mathcal{C}$ be a class of matroids closed under minors. A matroid M is an *excluded minor* for $\mathcal{C}$ if $M \notin \mathcal{C}$, but every proper minor of M belongs to $\mathcal{C}$.

The class of $\mathrm{GF}(p^r)$-representable matroids is closed under taking minors. Thus, it is natural to ask for minimal matroids which fail to be representable over $\mathrm{GF}(p^r)$.

Open Problem 1.1. Let p be a prime number. Let r be a positive integer. List the minimal excluded minors for the class of $\mathrm{GF}(p^r)$-representable matroids and prove that the given list is complete up to isomorphism.

For $p^r = 2$, the excluded minor list is known: a matroid is representable over $\mathrm{GF}(2)$ if and only if it has no minor isomorphic to $U_{2,4}$ [1]. For $p^r = 3$, there is also a finite explicit list: a matroid is representable over $\mathrm{GF}(3)$ if and only if it has no minor isomorphic to $U_{2,5}$, $U_{3,5}$, F_7, or F_7^* [2, 3]. In general, Rota's excluded-minor conjecture, proven by Geelen, Gerards and Whittle, says that for every finite field $\mathrm{GF}(p^r)$ there are only finitely many excluded minors for the class of $\mathrm{GF}(p^r)$-representable matroids [4, 5].

We have the following parametrized version of the open problem.

Open Problem 1.2. Let r be a positive integer. If there is a prime number p and a positive integer m such that $r = p^m$, then list the minimal excluded minors for the class of $\mathrm{GF}(p^m)$-representable matroids and prove that the list is complete up to isomorphism.

In the formal statement, the list is given as a finite set L of matroids together with its cardinality k: the members of L are pairwise non-isomorphic excluded minors, and every finite matroid that is not $\mathrm{GF}(p^m)$-representable has a minor isomorphic to a member of L. This rules out indirect solutions that prove the existence of a complete list without exhibiting one.

References

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